



Limits: RACE-01

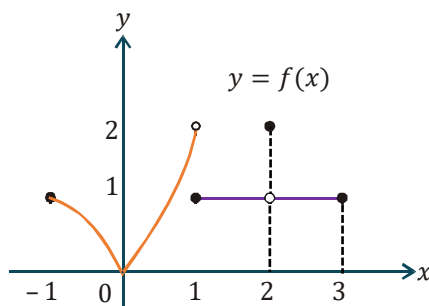
Single Correct Type Question

1. $\lim_{x \rightarrow 2} \frac{x^3 - 6x^2 + 11x - 6}{x^2 - 6x + 8}$ is equal to
 (A) $\frac{3}{2}$ (B) $\frac{1}{2}$
 (C) 1 (D) Does not exist
2. $\lim_{x \rightarrow 3} \frac{x^3 - 7x^2 + 15x - 9}{x^4 - 5x^3 + 27x - 27}$ is equal to
 (A) $2/9$ (B) $1/9$
 (C) $-2/9$ (D) Does not exist
3. $\lim_{x \rightarrow 1} \left(\frac{1}{1-x} - \frac{3}{1-x^3} \right)$ is equal to
 (A) -1 (B) 0
 (C) 1 (D) Does not exist
4. $\lim_{x \rightarrow 1} \frac{\left(\sum_{k=1}^{100} x^k \right) - 100}{x - 1} = \underline{\hspace{2cm}}$
 (A) 5000 (B) 5050
 (C) 5005 (D) 10100
5. $\lim_{x \rightarrow 2} \frac{\sqrt{1 + \sqrt{2+x}} - \sqrt{3}}{x - 2}$ is equal to
 (A) $\frac{1}{\sqrt{3}}$ (B) $\sqrt{3}$
 (C) $\frac{1}{4\sqrt{3}}$ (D) $\frac{1}{8\sqrt{3}}$
6. $\lim_{n \rightarrow \infty} (n - \sqrt{n^2 + n})$ is equal to
 (A) 1 (B) $\frac{1}{2}$
 (C) $-\frac{1}{2}$ (D) does not exist
7. The value of $\lim_{n \rightarrow \infty} \frac{3(n+1)}{(n+1)^3 \lfloor 3n \rfloor}, n \in N$ is _____
 (A) 27 (B) 9
 (C) 3 (D) 81

Comprehension

Paragraph for Questions 8 and 9

Graph of $y = f(x)$ is given below. Answer the two questions based on it.



8. Which of the following are correct?
- (A) $\lim_{x \rightarrow 1^+} f(x) = 1$ (B) $\lim_{x \rightarrow 2} f(x) = 2$
- (C) $\lim_{x \rightarrow 1^-} f(x) = 2$ (D) $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x)$
9. Which of the following are incorrect?
- (A) $\lim_{x \rightarrow c} f(x)$ exists at every $c \in [-1, 1]$ (B) $\lim_{x \rightarrow c} f(x)$ exists at every $c \in (1, 3)$
- (C) $\lim_{x \rightarrow 1^-} f(x) = 0$ (D) $\lim_{x \rightarrow 3^+} f(x)$ exists

Subjective Type Questions:

10. Evaluate: $\lim_{x \rightarrow a} \frac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}}, a \neq 0$
11. If $G(x) = -\sqrt{25-x^2}$, then find the $\lim_{x \rightarrow 1} \left(\frac{G(x) - G(1)}{x-1} \right)$
12. Find $\lim_{n \rightarrow \infty} \left(1 - \frac{1}{2^2} \right) \left(1 - \frac{1}{3^2} \right) \left(1 - \frac{1}{4^2} \right) \dots \left(1 - \frac{1}{n^2} \right)$
13. Evaluate: $\lim_{n \rightarrow \infty} \frac{1 \cdot n + 2 \cdot (n-1) + 3 \cdot (n-2) \dots + n \cdot 1}{n^3}$

Answer Key

Question	1	2	3	4	5	6	7	8	9
Answer	B	A	A	B	D	C	A	A,C,D	C,D

10. $\frac{2}{3\sqrt{3}}$

11. $\frac{1}{\sqrt{24}}$

12. $\frac{1}{2}$

13. $\frac{1}{6}$



Limits: RACE-02

Single Correct Type Question

1. If the value of $\lim_{n \rightarrow \infty} n^3 \left(1 - \frac{3}{7}\right) \left(1 - \frac{3}{8}\right) \left(1 - \frac{3}{9}\right) \dots \left(1 - \frac{3}{n}\right)$ is k , then sum of the digits of k is?
 (A) 6 (B) 2 (C) 3 (D) 8
2. $\lim_{x \rightarrow 0} \frac{(1 - \cos 2x) \sin 5x}{x^2 \sin 3x} =$
 (A) $10/3$ (B) $3/10$ (C) $6/5$ (D) $5/6$
3. $\lim_{x \rightarrow 1^-} \frac{1 - \sqrt{x}}{(\cos^{-1} x)^2}$ is equal to
 (A) $\frac{1}{2}$ (B) $-\frac{1}{4}$ (C) $-\frac{1}{2}$ (D) $\frac{1}{4}$
4. $\lim_{x \rightarrow 0} \frac{64}{x^8} \left[1 - \cos \frac{x^2}{2} - \cos \frac{x^2}{4} + \cos \frac{x^2}{2} \cos \frac{x^2}{4} \right] =$ _____
 (A) 0.25 (B) 0.50 (C) 0.75 (D) 1
5. $\lim_{x \rightarrow 0} \frac{e^{x^2} - \cos x}{x^2} =$ _____
 (A) $\frac{3}{2}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $e - 1$

Matrix Match Type Questions

6. Column-I

Column-II

(P) $\lim_{x \rightarrow 0} \frac{2^x - 1}{(1+x)^{1/2} - 1}$

(1) 2

(Q) $\lim_{x \rightarrow 0} \frac{e^{3x} - 1}{x/2}$

(2) $2 \ln 2$

(R) $\lim_{x \rightarrow 0} \frac{e^{\sin 2x} - e^{\sin x}}{x}$

(3) 6

(S) $\lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2}{x^2}$

(4) 1

- (A) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$
 (B) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$
 (C) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 2; S \rightarrow 4$
 (D) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 4$

Integer Type Question

7. The value of $\lim_{x \rightarrow \frac{\pi}{4}} \frac{1 - \tan x}{1 - \sqrt{2} \sin x}$ is?
8. $\lim_{n \rightarrow \infty} \frac{(n+1)^4 - (n-1)^4}{(n+1)^4 + (n-1)^4}$ is equal to
9. Using Sandwich theorem, evaluate

$$\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n^2}} + \frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \dots + \frac{1}{\sqrt{n^2+2n}} \right)$$

Subjective Type Question

10. Evaluate $\lim_{x \rightarrow 0} \left[\frac{-2x}{\tan x} \right]$, where $[.]$ denotes greatest integer function.
11. Evaluate $\lim_{x \rightarrow y} \frac{\sin^2 x - \sin^2 y}{x^2 - y^2}$
12. Evaluate $\lim_{h \rightarrow 0} \frac{(a+h)^2 \sin(a+h) - a^2 \sin a}{h}$

Answer Key

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	A	D	A	A	D	2	0	2.00	-2

11. $\left(\frac{\sin 2y}{2y} \right)$ 12. $a^2 \cos a + 2a \sin a$



Limits: RACE-03

Single Correct Type Question

- If $f(x) = \left(\frac{x^2 + 5x + 3}{x^2 + x + 2} \right)^x$ then value of $\lim_{x \rightarrow \infty} f(x)$ is
 (A) e^2 (B) e^4 (C) e^6 (D) e
- $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{\frac{\sin x}{x}}{1 - \frac{\sin x}{x}}}$ is equal to
 (A) e^{-1} (B) 1 (C) 0 (D) e^2
- $\lim_{x \rightarrow 0} (1+x)^{\cos ec x}$ is equal to
 (A) e (B) e^2 (C) e^{-1} (D) 1
- If x_1 and x_2 are the real and distinct roots of $ax^2 + bx + c = 0$, then value of $\lim_{x \rightarrow x_1} (1 + \sin(ax^2 + bx + c))^{\frac{1}{x-x_1}}$ is
 (A) $e^{a(x_1+x_2)}$ (B) $e^{(x_1-x_2)}$ (C) e^a (D) $e^{a(x_1-x_2)}$
- $\lim_{x \rightarrow 0} \left(\frac{a^x + b^x + c^x}{3} \right)^{2/x}$; $(a, b, c > 0)$ is equal to
 (A) $(abc)^{\frac{1}{3}}$ (B) $\frac{1}{3}$ (C) $(abc)^{\frac{2}{3}}$ (D) $\frac{(abc)^{\frac{2}{3}}}{3}$
- $\lim_{x \rightarrow 0} (1-2x)^{1/x}$ is equal to
 (A) e^{-2} (B) e (C) e^2 (D) e^{-1}
- $\lim_{x \rightarrow 1} x^{\cot \pi x}$ is equal to
 (A) $e^{\frac{2}{\pi}}$ (B) e (C) 1 (D) $e^{\frac{1}{\pi}}$

Multiple Correct Questions

8. Which of the following is correct

(A) $\lim_{x \rightarrow \infty} x \{ \ell n(x+a) - \ell nx \} = e^a$ (B) $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^{pn+q} = e^{p-q}$

(C) $\lim_{x \rightarrow 0} (1 + \tan^2 \sqrt{x})^{\frac{1}{2x}} = e^{\frac{1}{2}}$ (D) $\lim_{x \rightarrow \infty} \left(\frac{x+6}{x+1} \right)^{x+4} = e^5$

Limits

Integer Type Questions

9. $\lim_{x \rightarrow \infty} \left(\frac{1+5x^2}{1+3x^2} \right)^{-x^2}$ is equal to
10. $\lim_{n \rightarrow \infty} (4^n + 5^n)^{1/n}$ is equal to
11. $\lim_{x \rightarrow \infty} \left(\frac{x^2 + x - 1}{3x^2 + 2x + 4} \right)^{\frac{3x^2 + x}{x-2}}$ is equal to
12. $\lim_{x \rightarrow 0} (\cos x)^{\cot x}$ is equal to
13. If $\lim_{x \rightarrow \infty} \left[\frac{2x^2 + 3}{2x^2 + 5} \right]^{8x^2 + 3} = e^{-k}$, then the value of k is

Answer Key

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	A	A	D	C	A	D	C,D	0	5
Question	11	12	13							
Answer	0	1	8							



Limits: RACE-04

Single Correct Type Question

1. $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$ is equal to
 (A) $\frac{1}{2}$ (B) 2
 (C) e (D) 4
2. $\lim_{x \rightarrow 0} \frac{\sin x - x + \frac{x^3}{6}}{x^5}$ is equal to
 (A) $\frac{1}{5}$ (B) $\frac{1}{30}$
 (C) $\frac{1}{120}$ (D) 0
3. The value of $\lim_{h \rightarrow 0} \frac{\log_e(1+2h) - 2\log_e(1+h)}{h^2}$ is
 (A) 0 (B) -1
 (C) 1 (D) does not exist
4. The value of $\lim_{x \rightarrow 0} \frac{e^x \sin x - x - x^2}{x^3}$ is
 (A) 1 (B) 0
 (C) $1/3$ (D) $2/3$
5. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x \log(1+x)}$ is equal to
 (A) $\frac{1}{2}$ (B) $\frac{1}{3}$
 (C) 0 (D) 1
6. $\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x} \right)$ is equal to
 (A) $1/3$ (B) $1/6$
 (C) $-1/3$ (D) $-1/6$
7. $\lim_{x \rightarrow 0} \frac{(1+x)^n - 1}{x} =$ _____
 (A) n (B) e
 (C) 0 (D) 1

Matrix Match Type Questions

8. Column-I

Column-II

(P) $\lim_{x \rightarrow 0} \frac{x - \sin x}{\sin(x^3)}$

(1) $-\frac{1}{2}$

(Q) $\lim_{x \rightarrow 0} \frac{x - \tan^{-1} x}{x^3}$

(2) $-\frac{1}{6}$

(R) $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$

(3) $\frac{1}{6}$

(S) $\lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x + \log(1 - x)}{x^3}$

(4) $\frac{1}{3}$

(A) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$

(B) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$

(C) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 2; S \rightarrow 4$

(D) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$

Answer Key

Question	1	2	3	4	5	6	7	8
Answer	A	C	B	C	A	C	A	A